Cascaded $\chi^{(2)}{:}\chi^{(2)} \to \chi^{(3)}_{\text{eff}}$ in a Doubly-Resonant Metasurface

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1 Conventions

Real fields written as sums of complex envelopes (Boyd's convention, no prefactor $\frac{1}{2}$),

$$\mathcal{E}_i(\mathbf{r},t) = \sum_n \big[E_{n,i}(\mathbf{r},t)\, e^{-i\omega_n t} + \text{c.c.}\big], \qquad \mathcal{P}_i(\mathbf{r},t) = \sum_n \big[P_{n,i}(\mathbf{r},t)\, e^{-i\omega_n t} + \text{c.c.}\big], \tag{1}$$

with $E_{n,i}, P_{n,i}$ slowly varying. The second-order polarisation

$$P_i^{(2)}(\omega_\sigma) = \varepsilon_0 \sum_{(p,q)} \chi^{(2)}_{ijk}(\omega_\sigma;\omega_p,\omega_q)\, E_j(\omega_p)\, E_k(\omega_q), \tag{2}$$

gives for the two relevant channels

$$P_i^{(2)}(2\omega_1) = \varepsilon_0\, \chi^{(2)}_{ijk}(2\omega_1;\omega_1,\omega_1)\, E_j(\omega_1)\, E_k(\omega_1) \qquad \text{(SHG)} \tag{3}$$

$$P_i^{(2)}(\omega_1) = 2\varepsilon_0\, \chi^{(2)}_{ijk}(\omega_1;2\omega_1,-\omega_1)\, E_j(2\omega_1)\, E_k^*(\omega_1) \qquad \text{(DFG)} \tag{4}$$

The factor 2 counts orderings $(2\omega_1,-\omega_1)$ and $(-\omega_1,2\omega_1)$. The native Kerr response is

$$P_i^{(3)}(\omega_1) = 3\varepsilon_0\, \chi^{(3)}_{ijkl}(\omega_1;\omega_1,\omega_1,-\omega_1)\, E_j(\omega_1) E_k(\omega_1) E_l^*(\omega_1). \tag{5}$$

Under Kleinman symmetry

$$\chi^{(2)}_{ijk}(2\omega_1;\omega_1,\omega_1) = \chi^{(2)}_{kij}(\omega_1;2\omega_1,-\omega_1) \equiv \chi^{(2)}_{ijk}(\mathbf{r}). \tag{6}$$

Scalar $\chi^{(2)}$. The symbol $\chi^{(2)}_{ijk}(\mathbf{r})$ *with* indices is always the full position-dependent rank-3 tensor introduced above. The *unsubscripted* symbol $\chi^{(2)}$ (and its modulus $|\chi^{(2)}|$) is a single scalar magnitude characterising the active component of that tensor inside the nonlinear region $V_{\text{NL}} \subseteq V_{\text{eff}}$:

$$\boxed{\,|\chi^{(2)}| \;\equiv\; \max_{(ijk),\,\mathbf{r}\in V_{\text{NL}}} \big|\chi^{(2)}_{ijk}(\mathbf{r})\big|.\,} \tag{7}$$

For a homogeneous host with a single dominant tensor element, $|\chi^{(2)}|$ is just that element (e.g. $|\chi^{(2)}| = 2d_{33}$ in $LiNbO_3$ with the z-polarised eigenmodes; $|\chi^{(2)}| = 2d_{14}$ in GaAs/GaP/AlGaAs; etc.). In the bulk SVEA expressions of §3(a), $\chi^{(2)}$ denotes the same scalar projected onto the plane-wave polarisations, taken real and positive by an overall choice of phase.

This scalar is the dimensional prefactor that we pull out of the tensorial integrals to make the geometric coefficients $\widetilde{\eta}$ and η_{geom} in §5 dimensionless. With this convention, $|\chi^{(2)}|^2$ in (15) and (34) carries the material units, and all mode-shape, fill-fraction, and tensor-projection information sits in η_{geom} with $0 \le \eta_{\text{geom}} \le 1$.

2 Modes and energy normalisation

Linear eigenmodes $\mathbf{f}_n(\mathbf{r}), \mathbf{h}_n(\mathbf{r})$ satisfy

$$\nabla\times\nabla\times\mathbf{f}_n(\mathbf{r}) = \frac{\omega_n^2}{c^2}\,\varepsilon_r(\mathbf{r},\omega_n)\,\mathbf{f}_n(\mathbf{r}), \tag{8}$$

with $\mathbf{h}_n = -i\nabla\times\mathbf{f}_n/(\omega_n\mu_0)$. Brillouin energy normalisation:

$$\int\!\big[\,\varepsilon_0\,\widetilde{\varepsilon}_n(\mathbf{r})\,|\mathbf{f}_n|^2 + \mu_0|\mathbf{h}_n|^2\,\big]\,d^3r = 1, \qquad \widetilde{\varepsilon}_n(\mathbf{r}) \equiv \left.\frac{\partial(\omega\,\varepsilon_r(\mathbf{r},\omega))}{\partial\omega}\right|_{\omega=\omega_n}. \tag{9}$$

Modal amplitude $a_n(t)$ defined by $E_{n,i}(\mathbf{r},t) = a_n(t)\,f_{n,i}(\mathbf{r})$; time-averaged energy in mode n is $|a_n|^2$. Projection norm $N_n \equiv \frac{1}{2}\int[\varepsilon_0\widetilde{\varepsilon}_n|\mathbf{f}_n|^2+\mu_0|\mathbf{h}_n|^2]\,d^3r$ ($=\int\varepsilon_0\varepsilon_r|\mathbf{f}_n|^2$ for non-dispersive media; $=\frac{1}{2}$ here); in general the stored energy is $2N_n|a_n|^2$.

3 Cascaded polarisation from $\chi^{(2)}\cdot G\cdot\chi^{(2)}$

The SH field at leading order solves

$$\big[\nabla\times\nabla\times\,-k_2^2\,\varepsilon_r(\mathbf{r},2\omega_1)\big]\mathbf{E}(2\omega_1) = k_2^2\,\mu_0 c^2\,\mathbf{P}^{\mathrm{NL}}(2\omega_1), \qquad k_2\equiv 2\omega_1/c, \tag{10}$$

with dyadic Green's function $\overline{\overline{G}}(\mathbf{r},\mathbf{r}';2\omega_1)$:

$$E_j(\mathbf{r},2\omega_1) = k_2^2\int d^3r'\;G_{jl}(\mathbf{r},\mathbf{r}';2\omega_1)\,\chi^{(2)}_{lmn}(\mathbf{r}')\,E_m(\mathbf{r}',\omega_1)\,E_n(\mathbf{r}',\omega_1). \tag{11}$$

Substituting into the DFG polarisation (4),

$$P_i^{\mathrm{casc}}(\mathbf{r},\omega_1) = 2\varepsilon_0\,k_2^2\int d^3r'\;\underbrace{\chi^{(2)}_{ijk}(\mathbf{r})\,G_{jl}(\mathbf{r},\mathbf{r}';2\omega_1)\,\chi^{(2)}_{lmn}(\mathbf{r}')}_{\equiv\,\Xi^{\mathrm{casc}}_{ikmn}(\mathbf{r},\mathbf{r}')}\,E_m(\mathbf{r}',\omega_1)\,E_n(\mathbf{r}',\omega_1)\,E_k^*(\mathbf{r},\omega_1). \tag{12}$$

(a) Bulk, phase-mismatched (SVEA)

For plane waves $E(\omega_n,z) = A_n(z)\,e^{ik_nz}$, (10) with (3) as source yields

$$\frac{dA_2}{dz} = \frac{i\,\omega_2}{2\,n_2\,c}\,\chi^{(2)}\,A_1^2\,e^{-i\Delta kz}, \qquad \Delta k = k_2 - 2k_1. \tag{13}$$

For undepleted pump and $A_2(0)=0$,

$$A_2(z) = \frac{\omega_2\,\chi^{(2)}\,A_1^2}{2\,n_2\,c\,\Delta k}\big(1-e^{-i\Delta kz}\big). \tag{14}$$

Substituting into (4) and projecting back into the SVEA equation at ω_1, $dA_1/dz = (i\omega_1/(2n_1c))\,P^{\mathrm{NL}}(\omega_1)/\varepsilon_0\cdot e^{-ik_1z}$, produces $dA_1/dz \supset iK_{\mathrm{casc}}|A_1|^2A_1$ with

$$K_{\mathrm{casc}} = \frac{\omega_1\,\omega_2}{2n_1n_2c^2\,\Delta k}|\chi^{(2)}|^2\times(\text{phase factor}).$$

Matching to $dA_1/dz = i(3\omega_1/(2n_1c))\,\chi^{(3)}_{\mathrm{casc}}|A_1|^2A_1$:

$$\boxed{\chi^{(3)}_{\mathrm{casc,\,bulk}}(\omega_1) = -\frac{2\,\omega_1}{3\,c\,n_2\,\Delta k}\,|\chi^{(2)}|^2\,\mathcal{F}(\Delta k\,L),} \tag{15}$$

with $\mathcal{F}\to 1$ for $\Delta kL\gg 1$ and $\mathcal{F}\to\Delta kL/2$ for $\Delta kL\ll 1$.

(b) Single SH resonance (QNM)

Near an isolated SH quasi-normal mode,

$$G_{jl}(\mathbf{r},\mathbf{r}';2\omega_1) = -\frac{\varepsilon_0 c^2}{2\omega_2 N_2}\,\frac{f_{2,j}(\mathbf{r})\,f^*_{2,l}(\mathbf{r}')}{\Delta\omega + i\gamma_2/2} + (\text{bg}), \qquad \Delta\omega = 2\omega_1 - \omega_2. \tag{16}$$

With $E_m(\mathbf{r},\omega_1) = a_1\,f_{1,m}(\mathbf{r})$, define

$$M \equiv \int \chi^{(2)}_{ijk}(\mathbf{r})\,f^*_{2,i}\,f_{1,j}\,f_{1,k}\;d^3r, \qquad \overline{M} \equiv \int \chi^{(2)}_{ijk}(\mathbf{r})\,f^*_{1,i}\,f_{2,j}\,f^*_{1,k}\;d^3r = M^*. \tag{17}$$

Substituting (16) into (12) gives

$$P^{\text{casc}}_i(\mathbf{r},\omega_1) = -\frac{2\varepsilon_0^2\omega_1}{N_2}\,\frac{M\,|a_1|^2\,a_1}{\Delta\omega + i\gamma_2/2}\,\chi^{(2)}_{ijk}(\mathbf{r})\,f_{2,j}(\mathbf{r})\,f^*_{1,k}(\mathbf{r}). \tag{18}$$

Projecting onto $f^*_{1,i}$ and equating to the native-Kerr projection $\int f^*_{1,i}P^{\text{nat}}_i\,d^3r = 3\varepsilon_0\,\chi^{(3)}_{\text{casc}}\,|a_1|^2 a_1 \int |\mathbf{f}_1|^4\,d^3r$:

$$\boxed{\chi^{(3)}_{\text{casc}} = -\,\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2}\cdot\frac{|M|^2}{(\Delta\omega + i\gamma_2/2)\int|\mathbf{f}_1|^4\,d^3r}} \tag{19}$$

4 Coupled-mode equations

Projecting Maxwell's equations onto $\mathbf{f}^*_n$ (slowly-varying envelope) with phenomenological loss rates:

$$\frac{da_2}{dt} = \big(i\Delta\omega - \tfrac{\gamma_2}{2}\big)a_2 + i\beta\,a_1^2 \tag{20}$$

$$\frac{da_1}{dt} = -\tfrac{\gamma_1}{2}a_1 + i\beta^*\,a_1^*\,a_2 + i\alpha_3\,|a_1|^2 a_1 + \sqrt{\gamma_1^{(\text{rad})}}\,s_+ \tag{21}$$

with $s_- = -s_+ + \sqrt{\gamma_1^{(\text{rad})}}\,a_1$, $\Delta\omega \equiv 2\omega_1 - \omega_2$, $\gamma_1 = \gamma_1^{(\text{rad})} + \gamma_1^{(\text{nr})}$.

Manley–Rowe. In the lossless limit, differentiating $|a_n|^2$ from (20)–(21):

$$\frac{d|a_2|^2}{dt} = 2\,\text{Im}[\beta\,a_1^{*2}a_2], \qquad \frac{d|a_1|^2}{dt} = -2\,\text{Im}[\beta^{(1)}\,a_1^{*2}a_2], \tag{22}$$

$$\frac{d}{dt}(|a_1|^2 + |a_2|^2) = 2\,\text{Im}[(\beta - \beta^{(1)})\,a_1^{*2}a_2], \tag{23}$$

which vanishes $\forall a_1, a_2$ iff $\beta^{(1)} = \beta$.

Modal coefficients. With $\mathcal{E} = a(t)\mathbf{f}_n e^{-i\Omega t} + \text{c.c.}$ and a nonlinear polarisation $\mathbf{P}e^{-i\Omega t} + \text{c.c.}$, the slowly-varying projection of the wave equation onto $\mathbf{f}^*_n$ is

$$\dot{a} = i(\Omega - \omega_n)\,a + \frac{i\Omega}{2N_n}\int \mathbf{f}^*_n\cdot\mathbf{P}\;d^3r \tag{24}$$

(the first-order perturbation formula $\delta\omega = -\frac{\omega}{2}\int \mathbf{E}^*\cdot\delta\mathbf{P}/\int\varepsilon_0\varepsilon_r|\mathbf{E}|^2$). The SHG source gives $\beta = \varepsilon_0\omega_1 M/N_2$; the DFG source (degeneracy 2) gives the same coupling in (21) for $N_1 = N_2$; the SPM source (degeneracy 3) gives α_3. With $N_n = \frac{1}{2}$:

$$\boxed{\beta = \frac{\varepsilon_0\,\omega_1}{N_2}\,M = 2\,\varepsilon_0\,\omega_1\int \chi^{(2)}_{ijk}(\mathbf{r})\,f^*_{2,i}\,f_{1,j}\,f_{1,k}\;d^3r} \tag{25}$$

$$\boxed{\alpha_3 = \frac{3\,\varepsilon_0\,\omega_1}{2N_1}\int \chi^{(3)}_{ijkl}(\mathbf{r})\, f^*_{1,i}\, f_{1,j}\, f_{1,k}\, f^*_{1,l}\; d^3r} \tag{26}$$

(An earlier version used $\varepsilon_0\omega_1/4$ and $3\varepsilon_0\omega_1/8$ with $N_n = 1$, each $4\times$ too small; $\chi^{(3)}_{\rm casc} \propto |\beta|^2/\alpha_3$ was therefore $4\times$ too small.)

5 Adiabatic elimination ⇒ effective $\chi^{(3)}$

Assume $|\Delta\omega| \gg \gamma_2/2$, $|\dot a_2| \ll |\Delta\omega a_2|$. Setting $da_2/dt \to 0$:

$$a_2 = \frac{-\beta\, a_1^2}{\Delta\omega + i\gamma_2/2} \simeq \frac{-\beta\, a_1^2}{\Delta\omega}\left(1 - \frac{i\gamma_2}{2\Delta\omega}\right). \tag{27}$$

Substituting into (21):

$$\frac{da_1}{dt} = \left[-\tfrac{\gamma_1}{2} - \tfrac{|\beta|^2\gamma_2}{2\Delta\omega^2}|a_1|^2\right]a_1 + i\left[\alpha_3 - \tfrac{|\beta|^2}{\Delta\omega}\right]|a_1|^2 a_1 + \sqrt{\gamma_1^{(\rm rad)}}\, s_+. \tag{28}$$

Effective Kerr.

$$\boxed{\alpha_3^{\rm eff} = \alpha_3 - \frac{|\beta|^2}{\Delta\omega}} \tag{29}$$

Two-photon-like loss.

$$\gamma_3^{\rm eff} = \frac{|\beta|^2\,\gamma_2}{2\,\Delta\omega^2}, \qquad \gamma_3^{\rm eff}(\Delta\omega) = \frac{|\beta|^2(\gamma_2/2)}{\Delta\omega^2 + (\gamma_2/2)^2}. \tag{30}$$

Cascaded $\chi^{(3)}$. Inverting $\alpha_3 = (3\varepsilon_0\omega_1/2N_1)\,\chi^{(3)}_{\rm eff}\int |\mathbf{f}_1|^4\, d^3r$ with the cascaded coefficient $-\varepsilon_0^2\omega_1^2|M|^2/(N_1N_2\Delta\omega)$:

$$\boxed{\chi^{(3)}_{\rm casc} = -\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2\,\Delta\omega}\,\frac{\left|\int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i}\, f_{1,j}\, f_{1,k}\, d^3r\right|^2}{\int |\mathbf{f}_1|^4\, d^3r}\,.} \tag{31}$$

Explicit n-dependence. Using $|\mathbf{f}_n| \sim 1/\sqrt{2\varepsilon_0 n_n^2 V_{\rm eff}}$:

$$\int |\mathbf{f}_1|^4\, d^3r \sim \frac{\zeta_1}{4\varepsilon_0^2\, n_1^4\, V_{\rm eff}}, \tag{32}$$

$$\left|\int \chi^{(2)}_{ijk} f^*_{2,i} f_{1,j} f_{1,k}\, d^3r\right|^2 \sim \frac{|\chi^{(2)}|^2\,\widetilde{\eta}^2}{8\varepsilon_0^3\, n_2^2\, n_1^4\, V_{\rm eff}}. \tag{33}$$

All n_1 powers cancel:

$$\boxed{\chi^{(3)}_{\rm casc} \simeq -\frac{2\,\omega_1}{3\,n_2^2\,\Delta\omega}\,|\chi^{(2)}|^2\,\eta_{\rm geom}, \quad \eta_{\rm geom} \equiv \widetilde{\eta}^2/\zeta_1} \tag{34}$$

Explicit form of $\eta_{\rm geom}$. Introduce dimensionless mode profiles $u_n(\mathbf{r})$ by stripping the dimensional scale from $\mathbf{f}_n$,

$$\mathbf{f}_n(\mathbf{r}) \equiv \frac{u_n(\mathbf{r})}{\sqrt{2\,\varepsilon_0\, n_n^2\, V_{\rm eff}}}, \qquad \int |u_n(\mathbf{r})|^2\, d^3r \;=\; V_{\rm eff}, \tag{35}$$

where $V_{\rm eff}$ is the common effective mode volume (the normalisation (9) is reproduced for non-magnetic media with $\widetilde{\varepsilon}_n \sim n_n^2$ inside the mode and electric/magnetic equipartition). The u_n are $O(1)$ where the mode is concentrated and vanish far from it. Inserting (35) into the integrals introduced for the scaling estimate,

$$\int |\mathbf{f}_1|^4 \, d^3r = \frac{1}{4\,\varepsilon_0^2\, n_1^4\, V_{\rm eff}^2} \int |u_1|^4 \, d^3r, \tag{36}$$

$$\left| \int \chi^{(2)}_{ijk}\, f^*_{2,i}\, f_{1,j}\, f_{1,k}\; d^3r \right|^2 = \frac{1}{8\,\varepsilon_0^3\, n_1^4\, n_2^2\, V_{\rm eff}^3} \left| \int \chi^{(2)}_{ijk}\, u^*_{2,i}\, u_{1,j}\, u_{1,k}\; d^3r \right|^2, \tag{37}$$

fixes the dimensionless geometric factors as

$$\boxed{\zeta_1 \;\equiv\; \frac{1}{V_{\rm eff}} \int |u_1(\mathbf{r})|^4 \, d^3r, \qquad \widetilde{\eta} \;\equiv\; \frac{1}{|\chi^{(2)}|\, V_{\rm eff}} \left| \int \chi^{(2)}_{ijk}(\mathbf{r})\, u^*_{2,i}(\mathbf{r})\, u_{1,j}(\mathbf{r})\, u_{1,k}(\mathbf{r})\; d^3r \right|.} \tag{38}$$

Combining the two yields the closed-form

$$\boxed{\eta_{\rm geom} \;=\; \frac{\widetilde{\eta}^{\,2}}{\zeta_1} \;=\; \frac{\left| \int \chi^{(2)}_{ijk}(\mathbf{r})\, u^*_{2,i}\, u_{1,j}\, u_{1,k}\; d^3r \right|^2}{|\chi^{(2)}|^2\, V_{\rm eff} \int |u_1(\mathbf{r})|^4 \, d^3r}.} \tag{39}$$

Equivalently, eliminating the dimensionless profiles in favour of the Brillouin-normalised $\mathbf{f}_n$,

$$\eta_{\rm geom} = \frac{\varepsilon_0\, n_2^2 \left| \int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i}\, f_{1,j}\, f_{1,k}\; d^3r \right|^2}{|\chi^{(2)}|^2\, N_2 \int |\mathbf{f}_1(\mathbf{r})|^4 \, d^3r}, \tag{40}$$

which back-substituted into (34) reproduces (31) exactly (the n_2^2 cancels the $1/n_2^2$ in the prefactor and the ε_0 restores the tensorial form).

Bounds and limits. By Cauchy–Schwarz on the integrand of (39), with $|\chi^{(2)}(\mathbf{r})| \le |\chi^{(2)}|$ on its support and $\int |u_2|^2 \, d^3r = V_{\rm eff}$,

$$\left| \int \chi^{(2)}_{ijk}\, u^*_{2,i}\, u_{1,j}\, u_{1,k}\; d^3r \right|^2 \;\le\; |\chi^{(2)}|^2\, V_{\rm eff} \int |u_1|^4 \, d^3r \quad\Longrightarrow\quad 0 \;\le\; \eta_{\rm geom} \;\le\; 1.$$

For a uniform $\chi^{(2)}$ filling a fraction $V_{\rm NL}/V_{\rm eff}$ of the mode with perfectly aligned, top-hat profiles inside the nonlinear region, $\widetilde{\eta}, \zeta_1 \sim V_{\rm NL}/V_{\rm eff}$ and hence $\eta_{\rm geom} \sim V_{\rm NL}/V_{\rm eff}$ — the nonlinear fill fraction weighted by the tensor projection. Realistic (I guess) BIC/qBIC $LiNbO_3$ designs give $\eta_{\rm geom} \approx 0.1$–$0.5$; a value $\eta_{\rm geom} \approx 0.3$ is used in the numerical estimates (in the presentation particularly).

6 Bandwidth–enhancement sum rule

For $|a_1|^2$ modulated at Ω, $\delta a_2 = i\beta\, \delta(a_1^2)/(-i(\Delta\omega - \Omega) + \gamma_2/2)$. The frequency-dependent cascaded Kerr:

$$\alpha_3^{\rm eff}(\Omega) = \frac{i\,|\beta|^2}{\gamma_2/2 + i(\Omega - \Delta\omega)}, \qquad |\alpha_3^{\rm eff}(\Omega)| = \frac{|\beta|^2}{\sqrt{(\gamma_2/2)^2 + (\Omega - \Delta\omega)^2}}. \tag{41}$$

$|\alpha_3^{\rm eff}(\Omega)|^2$ is Lorentzian, FWHM γ_2, peak $4|\beta|^4/\gamma_2^2$, area

$$\int d\Omega\, |\alpha_3^{\rm eff}(\Omega)|^2 = \frac{2\pi\,|\beta|^4}{\gamma_2}. \tag{42}$$

At $|\Delta\omega| \sim \gamma_2$, $|\alpha_3^{\rm eff}|_{\max} \sim |\beta|^2/\gamma_2$, $\Delta\Omega \sim \gamma_2$. With $\omega_1/\gamma_2 = Q_2/2$:

$$|\chi^{(3)}_{\rm casc}|_{\max} \sim \frac{Q_2}{3\,n_2^2}\,|\chi^{(2)}|^2\,\eta_{\rm geom}. \tag{43}$$

Multiplying by $\Delta f_{\rm usable} \sim \gamma_2/(2\pi)$:

$$\boxed{|\chi^{(3)}_{\rm casc}|_{\max} \times \Delta f_{\rm usable} \lesssim \frac{\omega_1\,|\chi^{(2)}|^2\,\eta_{\rm geom}}{3\pi\,n_2^2} \quad (Q_2\text{-independent}).} \tag{44}$$

7 Per-photon nonlinear phase

On-resonance pump, fully-radiative limit $\gamma_1^{(\rm rad)} = \gamma_1$: $|a_1|^2 = (4/\gamma_1)|s_+|^2$, $\phi_{\rm NL} = \alpha_3^{\rm eff}|a_1|^2 \cdot (2/\gamma_1)$.

$$\boxed{\frac{\phi_{\rm NL}}{|s_+|^2} = \frac{8\,\alpha_3^{\rm eff}}{\gamma_1^2}} \tag{45}$$

For Architecture A (native), $\alpha_3^{\rm eff,A} = \alpha_3 \propto \chi^{(3)}_{\rm native}$. For B (cascade),

$$\alpha_3^{\rm eff,B} = -\frac{|\beta|^2}{\Delta\omega + i\gamma_2/2} \propto \frac{|\chi^{(2)}|^2\eta_{\rm geom}}{n_2^2\Delta\omega} \propto Q_2. \tag{46}$$

Define

$$\boxed{R \equiv \frac{|\alpha_3^{\rm eff,B}|}{|\alpha_3^{\rm eff,A}|} = \frac{|\beta|^2}{|\alpha_3|\,|\Delta\omega + i\gamma_2/2|} \overset{\Delta\omega \gg \gamma_2}{\simeq} \frac{|\chi^{(3)}_{\rm casc}|}{|\chi^{(3)}_{\rm native}|}.} \tag{47}$$

8 Kerr-driven observables

SPM. Linear in $\alpha_3^{\rm eff}$:

$$\boxed{\frac{\phi_{\rm NL}^{\rm A+B}}{\phi_{\rm NL}^{\rm A}} = 1 + R.} \tag{48}$$

XPM. Same scaling, with $\alpha_3^{\rm XPM} = 2\alpha_3^{\rm SPM}$:

$$\boxed{\frac{\alpha_3^{\rm XPM,\,A+B}}{\alpha_3^{\rm XPM,\,A}} = 1 + R.} \tag{49}$$

Bistability / switching. Steady state of (21) at detuning δ_p:

$$\left[\tfrac{\gamma_1}{2} - i(\delta_p + \alpha_3^{\rm eff}|a_1|^2)\right] a_1 = \sqrt{\gamma_1^{\rm rad}}\,s_+. \tag{50}$$

Scaling $y = \alpha_3^{\rm eff}|a_1|^2/(\gamma_1/2)$, $x = \delta_p/(\gamma_1/2)$, $u = |\alpha_3^{\rm eff}||s_+|^2\gamma_1^{\rm rad}/(\gamma_1/2)^3$:

$$y[1 + (x - y)^2] = u. \tag{51}$$

Three roots iff

$$\mathrm{sign}(\delta_p) = -\mathrm{sign}(\alpha_3^{\rm eff}), \qquad |\delta_p| > \sqrt{3}\,\gamma_1/2. \tag{52}$$

Cusp threshold $|a_1|_c^2 = \sqrt{3}\gamma_1/(2|\alpha_3^{\rm eff}|)$:

$$|s_+|^2_{\rm bist} \propto \frac{\gamma_1^2}{|\alpha_3^{\rm eff}|}, \tag{53}$$

$$\boxed{\frac{|s_+|^{2,\,\mathrm{A+B}}_{\rm bist}}{|s_+|^{2,\,\mathrm{A}}_{\rm bist}} = \frac{E^{\mathrm{A+B}}_{\rm switch}}{E^{\mathrm{A}}_{\rm switch}} = \frac{1}{1+R}.} \tag{54}$$

FWM. Write $a_1 = a_p + a_s e^{-i\Omega t} + a_i e^{+i\Omega t}$ with $|a_s|, |a_i| \ll |a_p|$. Collecting the $e^{+i\Omega t}$ component of $|a_1|^2 a_1$ gives source $\mathcal{S}_i = i\alpha_3^{\rm eff} a_p^2 a_s^*$ and detuning renormalisation $2|a_p|^2 a_i$ (dropped below). Idler steady state:

$$a_i = \frac{i\,\alpha_3^{\rm eff}\,a_p^2\,a_s^*}{\gamma_1/2 - i\Omega} \simeq \frac{i\,\alpha_3^{\rm eff}\,a_p^2\,a_s^*}{\gamma_1/2} \quad (|\Omega| < \gamma_1/2). \tag{55}$$

Using $|s_-^{(i)}|^2 = \gamma_1^{\rm rad}|a_i|^2$ and $|a_s| = 2\sqrt{\gamma_1^{\rm rad}}|s_+^{(s)}|/\gamma_1$:

$$\eta_{\rm FWM} \equiv \frac{|s_-^{(i)}|^2}{|s_+^{(s)}|^2} \propto \frac{|\alpha_3^{\rm eff}|^2\,|a_p|^4}{(\gamma_1/2)^2}. \tag{56}$$

$$\boxed{\frac{\eta^{\mathrm{A+B}}_{\rm FWM}}{\eta^{\mathrm{A}}_{\rm FWM}} = (1+R)^2.} \tag{57}$$

9 Imaginary part: cascaded saturable absorber

From (19), writing $1/(\Delta\omega + i\gamma_2/2) = (\Delta\omega - i\gamma_2/2)/[\Delta\omega^2 + (\gamma_2/2)^2]$:

$$\mathrm{Im}\,\chi^{(3)}_{\rm casc}(\Delta\omega) = \frac{2\,\varepsilon_0\,\omega_1}{3\,N_2} \cdot \frac{|M|^2\,(\gamma_2/2)}{[\Delta\omega^2 + (\gamma_2/2)^2]\int |\mathbf{f}_1|^4\,d^3r}, \tag{58}$$

$$\gamma_3^{\rm eff}(\Delta\omega) = \frac{|\beta|^2(\gamma_2/2)}{\Delta\omega^2 + (\gamma_2/2)^2}, \qquad \gamma_3^{\rm eff}(0) = \frac{|\beta|^2}{\gamma_2/2} = \frac{2Q_2}{\omega_2}|\beta|^2. \tag{59}$$

At $\Delta\omega = 0$ (Lorentzian peak, FWHM γ_2):

$$\mathrm{Im}\,\chi^{(3)}_{\rm casc}\big|_{\Delta\omega=0} \sim \frac{2\,Q_2}{3\,n_2^2}\,|\chi^{(2)}|^2\,\eta_{\rm geom}. \tag{60}$$

10 Amplitude modulator: $\Delta\omega(t)$ transfer function

Linearise $\Delta\omega(t) = \Delta\omega_0 + \delta\Omega\cos(\Omega_{\rm mod}t)$ in the SH equation (20). The positive-frequency sideband of δa_2:

$$A_+(\Omega_{\rm mod}) = \frac{(\delta\Omega/2)\,a_{2,0}}{\gamma_2/2 + i(\Omega_{\rm mod} - \Delta\omega_0)}, \qquad a_{2,0} = \frac{-\beta\,a_1^2}{\Delta\omega_0 + i\gamma_2/2}. \tag{61}$$

$$\left|\frac{\delta a_2}{a_{2,0}}\right|_{\Omega_{\rm mod}} = \frac{\delta\Omega/2}{\sqrt{(\gamma_2/2)^2 + (\Omega_{\rm mod} - \Delta\omega_0)^2}}. \tag{62}$$

Peak at $\Omega_{\rm mod} = \Delta\omega_0$, DC at $\Omega_{\rm mod} = 0$:

$$\left|\frac{\delta a_2}{a_{2,0}}\right|_{\rm peak} = \frac{\delta\Omega}{\gamma_2}, \quad \left|\frac{\delta a_2}{a_{2,0}}\right|_{\rm DC} = \frac{\delta\Omega}{\sqrt{\Delta\omega_0^2 + (\gamma_2/2)^2}}. \tag{63}$$

AC-peak/DC enhancement:

$$\boxed{\frac{|\delta\alpha_3^{\rm eff}|_{\rm peak}}{|\delta\alpha_3^{\rm eff}|_{\rm DC}} = \frac{\sqrt{\Delta\omega_0^2 + (\gamma_2/2)^2}}{\gamma_2} \xrightarrow[\Delta\omega_0 \gg \gamma_2]{} \frac{|\Delta\omega_0|}{\gamma_2}.} \tag{64}$$

11 Sensor: $d\alpha_3^{\rm casc}/d\Delta\omega$

With $b \equiv \gamma_2/2$, $\alpha_3^{\rm casc}(\Delta\omega) = -|\beta|^2\Delta\omega/[\Delta\omega^2 + b^2]$:

$$\frac{d\alpha_3^{\rm casc}}{d\Delta\omega} = -|\beta|^2 \, \frac{b^2 - \Delta\omega^2}{[\Delta\omega^2 + b^2]^2}. \tag{65}$$

Extrema: global max at $\Delta\omega = 0$, local max at $|\Delta\omega| = b\sqrt{3}$ (factor 8 smaller).

$$\boxed{\left|\frac{d\alpha_3^{\rm casc}}{d\Delta\omega}\right|_{\Delta\omega=0} = \frac{|\beta|^2}{b^2} = \frac{4|\beta|^2}{\gamma_2^2} \propto Q_2^2.} \tag{66}$$

Using (45),

$$\left|\frac{d\phi_{\rm NL}}{d\Delta\omega}\right|_{\rm max} = \frac{32\,|s_+|^2\,|\beta|^2}{\gamma_1^2\,\gamma_2^2} \propto Q_1^2\,Q_2^2. \tag{67}$$

12 Principal result

$$\boxed{\chi^{(3)}_{\rm casc}(\omega_1) = -\frac{2\,\varepsilon_0\,\omega_1}{3\,N_2\,\Delta\omega}\,\frac{\left|\int \chi^{(2)}_{ijk}(\mathbf{r})\,f^*_{2,i}\,f_{1,j}\,f_{1,k}\,d^3r\right|^2}{\int |\mathbf{f}_1(\mathbf{r})|^4\,d^3r} \;\simeq\; -\frac{2\,\omega_1}{3\,n_2^2\,\Delta\omega}\,|\chi^{(2)}|^2\,\eta_{\rm geom}\,,} \tag{68}$$

valid for $\gamma_2/2 \ll |\Delta\omega| \ll \omega_1$, $|\beta a_1| \ll |\Delta\omega|$.